

Torin Chacko

Backend Developer & Systems Engineer

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Education

University of California, Los Angeles
B.S., Mathematics of Computation

Expected 2028
Los Angeles, CA

- Relevant coursework: CS33 Computer Architecture and Operating Systems in x86-64 ASM, CS32 Data Structures & Algorithms, Linear Algebra, Real Analysis, Complex Analysis, Statistical Models

Experience

Software Engineering Intern
County of Marin, Information Services & Technology (IST)

06/2025 – 08/2026
San Rafael, CA

- Updated out-of-date systems for the County in C#.
- Worked with the LAND division to update their platform and tools to allow for more user access and use.
- *[SQL Queries, .NET Framework, HTML, CSS]*

Software Engineer
Numberzoo

07/2026 – Present
[Lucas Valley, CA]

- Connected client data with custom-built chat applications
- Built AI-powered workflow automation and automated document creation for small businesses
- *[SQL, PostGres, C#, .NET, Node, AI Automation]*

Projects

x86-64 Emulator
C++20, CMake

- Building a from-scratch x86-64 emulator and debugger covering instruction semantics, machine-code decoding, ELF loading, virtual memory, and syscalls
- Implemented CPU state and flag-correctness (CF/PF/AF/ZF/SF/OF) verified against native hardware with a dependency-free unit-test suite

Compartmental Human Metabolic Model
Python

- Built an hour-by-hour, ten-subsystem simulation of human metabolism responding to diet and exercise inputs
- Enforced strict double-entry mass and energy conservation across every organ compartment and environmental sink

Garmin Field Log
Python, GitHub Actions, HTML

- Connected Garmin API with a serverless fitness dashboard: a scheduled GitHub Actions job pulls Garmin Connect data daily and a static GitHub Pages frontend renders it
- Created a backfill feature to pull sequentially all historical data from Garmin and update the website with the new data.

Skills

Languages C, C++, Python, x86-64 Assembly, SQL, Advanced Mathematics
Tools .NET, Node, Git, CMake, Linux, MySQL, GitHub Actions, LaTeX